

Customer Retention & Sales Analysis Report

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Tools Used: google colab ,SQL, Python (Pandas), Power BI

1. Project Objective

The purpose of this analysis is to understand:

- Customer behavior and retention trends
- Order and revenue patterns
- Repeat purchases and payment performance
- Customer cohort retention patterns over time

The analysis was conducted using SQL for data extraction and insights, Python for data transformation, and Power BI for interactive visualizations.

2. Key SQL-Based Insights

Order Status Distribution:

Order Status	Total Orders
Delivered	5,057
Pending	5,069
Shipped	4,874

Revenue is fairly stable with minor fluctuations. Highest revenue month: March 2025 - ₹64,379.46

Month	Revenue (₹)
Jan 2020	60,638.78
Feb 2020	61,461.48
...	...
Apr 2025	57,148.69

- **Insight: Sales recovered in early 2025 after a slight drop in late 2024.**

C. Top 5 Customers by Spend

Customer ID	Total Spent (₹)
6477	2,798.16
6679	2,246.56
5920	2,206.81
7932	2,194.54
9964	2,177.64

- **Insight: A small portion of customers contribute significantly to overall revenue – loyal high-value buyers.**

D. Orders Per Month

- Steady order volumes with seasonal spikes.

- Lowest order dip: Possibly due to external disruptions or reduced marketing efforts.

E. Repeat Customer Behavior

Customers with Repeat Orders:

- Total repeat customers: 4,402
- One-time customers: 2,932

Top Repeaters:

Customer ID	Orders
2633	8
9955	7

Monthly Repeat Order Growth:

Month	Repeat Orders
Jan 2020	3
Apr 2025	188

- **Insight: Strong growth in repeat buyers – indicates increasing customer loyalty over time.**

F. Payments Analysis

Payment Status Distribution:

Status	Count
Completed	4,991
Failed	5,003
Pending	5,006

Failure Analysis:

- Success Rate: Only 33.27%
- Total Failed Payment Amount: ₹1,273,618.69
- Top Failed Payment Methods:
 - Bank Transfer (1,758 failures)
 - Credit Card (1,666)
 - PayPal (1,579)
- Insight: Low success rate may impact both revenue and customer experience. High-priority fix needed.

3. Cohort Analysis – Retention Study

Objective:

To understand user retention by tracking cohorts based on their first purchase month, and observing behavior over the next 0 to 64 months.

Methodology:

In Python:

- Converted wide matrix into long format.
- Used `pandas.melt()` to transform from 65 columns to 3 fields:
 - `cohort_month`
 - `cohort_index` (month since first purchase)
 - `retention_value`

In Power BI:

- Created a Matrix visual:
 - Rows: cohort_month
 - Columns: cohort_index
 - Values: retention_value
- Applied conditional formatting (heatmap)

Key Cohort Insights:

Metric	Insight
Highest retention drop	Between month 0 → 1 across most cohorts
Strongest cohort	Jan 2020, with higher-than-average Month 3–6 retention
Avg. cohort lifespan	~3–4 months
Month with best growth	Early 2025
Poor-performing cohort	April 2020 – very low Month 2+ retention